Python for Network Engineers: Complete Lab Guide

From Zero to Network Automation — A Comprehensive, Multi-Source Reference

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Sources Cross-Referenced in This Guide:

Abbreviation	Source	Focus
LPTHW	*Learn Python the Hard Way* (Zed Shaw)	Fundamentals via repetition
PyNEng	*Python for Network Engineers* (Natasha Samoylenko)	Python + networking context
KByers	*Python Course* (Kirk Byers — python_course_mar26)	Security automation (Check Point)
Netmiko	*Netmiko Course* (Kirk Byers — netmiko_course)	SSH device automation

How to Use This Guide: Each chapter teaches a concept, shows code from multiple authors side by side, and ends with 5 lab tasks. All answers are in **Appendix A** at the end.

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Chapter 1: Environment Setup & First Steps

1.1 Why Python for Network Engineers?

Python has become the de facto language for network automation. Whether you manage Cisco IOS, NX-OS, Arista EOS, Juniper JunOS, or security platforms like Check Point, Python gives you a single language to automate them all.

What you will automate with Python:

- SSH into routers and switches to collect `show` commands
- Push configuration changes to hundreds of devices simultaneously
- Parse unstructured CLI output into structured data
- Interact with REST APIs (Cisco DNA Center, Check Point Management, Meraki, etc.)
- Generate configurations from templates (Jinja2)
- Build compliance audit scripts

1.2 Installing Python

On Linux (Ubuntu/Debian) — Recommended for Network Engineers

```bash
sudo apt update
sudo apt install -y python3 python3-pip python3-venv git
python3 --version
```

On macOS

```bash
# Install Homebrew first if not installed
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/i
brew install python3
```

On Windows

Download the installer from python.org. Check "Add Python to PATH" during installation.

1.3 Virtual Environments

> **PyNEng:** "Virtual environments allow different projects to have different sets of installed packages... this avoids conflicts between package versions."

> **LPTHW:** Focus on getting a clean terminal setup before writing any code.

```bash
# Create a virtual environment
python3 -m venv ~/netauto_venv

# Activate it
source ~/netauto_venv/bin/activate    # Linux/macOS
# netauto_venv\Scripts\activate       # Windows

# Verify
which python
pip install --upgrade pip
```

Installing Key Packages for Network Automation

```bash
pip install netmiko paramiko requests pyyaml jinja2 rich textfsm
pip install python-dotenv ipdb
```

1.4 Your First Python Script

LPTHW Approach — Type it, run it, see the output

Create a file called `ex1.py`:

```python
print("Hello, Network Engineer!")
print("I will automate your switches.")
print("I will automate your routers.")
print("I will automate your firewalls.")
```

Run it:

```bash
python3 ex1.py
```

PyNEng Approach — Interactive interpreter (IPython)

```bash
pip install ipython
ipython
```

```python
In [1]: print("Hello, Network Engineer!")
Hello, Network Engineer!

In [2]: 2 + 2
Out[2]: 4

In [3]: exit()
```

KByers Approach — Use the Rich library for pretty output

```python
#!/usr/bin/env python
from rich import print

print("Hello, Network Engineer!")
print("[bold green]Automation Ready![/bold green]")
```

1.5 The Python Interpreter and Comments

```python
# This is a comment — Python ignores this line
# Comments explain WHY, not WHAT

hostname = "core-rtr-01"  # inline comment

# Multi-line comments use consecutive # symbols
# or triple-quoted strings (docstrings)
"""
This script connects to all routers
and collects the running configuration.
"""
```

1.6 Git Basics (Version Control Your Scripts)

> **PyNEng:** "Git is a distributed version control system... even if you work alone, git can be useful."

```bash
```

```
# Initial setup
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"

# Create a repository
mkdir ~/network_automation
cd ~/network_automation
git init

# Basic workflow
git add script.py
git commit -m "Add initial device connection script"
git log --oneline
```

Chapter 1 Lab Tasks

Task 1.1: Install Python 3 on your system. Create a virtual environment called `netauto`. Activate it and install `netmiko`, `requests`, `pyyaml`, and `rich`. Take a screenshot of `pip list` output.

Task 1.2: Write a script called `my_info.py` that prints your name, your role (e.g., "Network Engineer"), and the number of devices you manage. Use at least one comment.

Task 1.3: Open the Python interactive interpreter (or IPython). Perform these calculations: `255 * 255`, `10 / 3`, `2 ** 32`. What data types does Python return for each?

Task 1.4: Initialize a git repository in a new folder. Create a file called `hello.py` with a print statement, add it to git, and commit with the message "first commit". Run `git log` and note the output.

Task 1.5: Create a script that uses the `rich` library to print "Router connected!" in bold green and "Firewall blocked!" in bold red. (Hint: `from rich import print`, use `[bold green]` tags.)

Chapter 2: Variables, Numbers, and Math

2.1 Variables

A variable stores a value. In Python, you do not declare variable types — Python infers them.

LPTHW Style — Exercise 4: Variables and Names

```python
```

```
cars = 100
space_in_a_car = 4.0
drivers = 30
passengers = 90
cars_not_driven = cars - drivers
cars_driven = drivers
carpool_capacity = cars_driven * space_in_a_car
average_passengers_per_car = passengers / cars_driven

print("There are", cars, "cars available.")
print("There are only", drivers, "drivers available.")
print("We need to put about", average_passengers_per_car, "in each car.")
```

PyNEng Style — Network Variables

python

```
In [1]: hostname = "core-rtr-01"

In [2]: interface_count = 48

In [3]: cpu_load = 23.7

In [4]: type(hostname)
Out[4]: str

In [5]: type(interface_count)
Out[5]: int

In [6]: type(cpu_load)
Out[6]: float
```

KByers Style — Networking Context

python

```python
#!/usr/bin/env python
from rich import print

fw_name = "chkpnt-pod99"
ip_address = "3.77.44.109"
os_version = "R82"

print(f"{fw_name=}")          # Python 3.8+ debug format
print(f"{ip_address=}")
print(f"{os_version=}")
```

Output:

```
fw_name='chkpnt-pod99'
ip_address='3.77.44.109'
os_version='R82'
```

2.2 Variable Naming Rules

```python
# GOOD names (descriptive, snake_case)
interface_name = "GigabitEthernet0/1"
vlan_id = 100
subnet_mask = "255.255.255.0"

# BAD names (avoid)
x = "GigabitEthernet0/1"    # too short, meaningless
InterfaceName = "Gi0/1"      # PascalCase is for classes
2nd_vlan = 200                # cannot start with a number — SyntaxError!
```

2.3 Numbers and Math

> **PyNEng:** "Python has several numeric types: int (integers) and float (floating-point numbers)."

```python
```

```python
# Integer math
vlans_per_switch = 4094
switches = 5
total_vlans = vlans_per_switch * switches
print(total_vlans)            # 20470

# Float math
bandwidth_gbps = 10.0
utilization = 0.73
used_bandwidth = bandwidth_gbps * utilization
print(used_bandwidth)         # 7.3

# Division always returns float
print(10 / 3)                 # 3.3333333333333335

# Floor division (integer result)
print(10 // 3)                # 3

# Modulo (remainder)
print(10 % 3)                 # 1

# Exponentiation
print(2 ** 32)                # 4294967296 (IPv4 address space)
print(2 ** 128)               # IPv6 address space

# Rounding
print(round(10/3, 2))         # 3.33
```

2.4 Type Conversions

```python
```

```python
# String to integer
vlan_str = "100"
vlan_int = int(vlan_str)
print(type(vlan_int))         # <class 'int'>

# Integer to string
port = 22
port_str = str(port)

# Float to integer (truncates, does NOT round)
print(int(3.9))               # 3

# Binary and Hexadecimal conversions (common in networking)
print(bin(255))               # '0b11111111'
print(hex(255))               # '0xff'
print(int('ff', 16))          # 255
print(int('11111111', 2))     # 255
print(bin(int('ff', 16)))     # '0b11111111'
```

2.5 Comparison Operators

```python
packets_sent = 1000
packets_received = 985

print(packets_sent > packets_received)     # True
print(packets_sent == packets_received)    # False
print(packets_sent != packets_received)    # True
print(packets_sent >= 1000)                # True

# Calculate packet loss
loss = packets_sent - packets_received
loss_pct = (loss / packets_sent) * 100
print(f"Packet loss: {loss_pct}%")         # Packet loss: 1.5%
```

Chapter 2 Lab Tasks

Task 2.1: Create variables for a network device: `hostname`, `ip_address`, `model`, `ios_version`, `uptime_days`. Print them all using f-strings.

Task 2.2: Calculate the total number of usable IP addresses in a /24, /16, and /8 subnet using the formula `2^(32-prefix) - 2`. Print each result.

Task 2.3: Convert the decimal number 192 to binary and hexadecimal. Then convert the hex value "C0A80001" to decimal. What IP address does that represent?

Task 2.4: You have 1000 packets sent and 987 received. Calculate the packet loss percentage, rounded to 2 decimal places. Store each step in a variable.

Task 2.5: A 10 Gbps link runs at 73% utilization. Calculate the used bandwidth in Mbps. Then calculate how many megabytes can be transferred in 60 seconds at that rate (1 byte = 8 bits).

Chapter 3: Strings and Text Processing

Strings are the most common data type in network automation. IP addresses, hostnames, show command output — it is all strings.

3.1 Creating Strings

```python
# Three ways to create strings
hostname = "core-rtr-01"
interface = 'GigabitEthernet0/0/0'
multiline_config = """interface Loopback0
 ip address 10.0.0.1 255.255.255.255
 no shutdown"""

print(hostname)
print(interface)
print(multiline_config)
```

3.2 String Operations

```python
```

```python
hostname = "core-rtr-01"

# Length
print(len(hostname))          # 11

# Concatenation
domain = "example.com"
fqdn = hostname + "." + domain
print(fqdn)                   # core-rtr-01.example.com

# Repetition
divider = "-" * 40
print(divider)                # ----------------------------------------

# Membership test
print("rtr" in hostname)      # True
print("sw" in hostname)       # False
```

3.3 String Methods (Essential for Network Engineers)

LPTHW + PyNEng Combined

```python
# .split() — Break strings apart (CRITICAL for parsing CLI output)
ip_address = "198.51.100.0/24"
network, mask = ip_address.split("/")
print(f"Network: {network}")     # Network: 198.51.100.0
print(f"Mask: {mask}")           # Mask: 24

octets = network.split(".")
print(octets)                    # ['198', '51', '100', '0']

# .join() — Combine list into string
commands = ['interface Gi0/1', 'switchport mode access', 'switchport access vlan 10'
config_block = "\n".join(commands)
print(config_block)
```

KByers Approach — IPv4 and IPv6 Parsing

```python
```

```python
#!/usr/bin/env python
from rich import print

ip_address = "198.51.100.0/24"
ipv6_address = "2001:db8:1:1::/64"

# IPv4 parsing
ipv4_network, ipv4_mask = ip_address.split("/")
octet1, octet2, octet3, octet4 = ipv4_network.split(".")

divider = "-" * 15
print(f"{'octet1':^15} {'octet2':^15} {'octet3':^15} {'octet4':^15}")
print(f"{divider:15} {divider:15} {divider:15} {divider:15}")
print(f"{octet1:^15} {octet2:^15} {octet3:^15} {octet4:^15}")

# IPv6 parsing
ipv6_network, ipv6_mask = ipv6_address.split("/")
start_ipv6, _ = ipv6_network.split("::")
hextet1, hextet2, hextet3, hextet4 = start_ipv6.split(":")
print(f"Hextets: {hextet1}, {hextet2}, {hextet3}, {hextet4}")
```

3.4 More Essential String Methods

python

```python
line = "  Interface GigabitEthernet0/1 is up, line protocol is up  "

# .strip() — Remove whitespace (critical for file/CLI output parsing)
print(line.strip())

# .upper() / .lower()
print("vlan100".upper())         # VLAN100
print("OSPF".lower())            # ospf

# .startswith() / .endswith()
if line.strip().startswith("Interface"):
    print("This is an interface line")

# .replace()
new_line = line.replace("up", "down")
print(new_line)

# .count()
config = "permit ip any any\npermit tcp any any\ndeny ip any any"
print(config.count("permit"))    # 2

# .find() — returns index, or -1 if not found
print(line.find("up"))           # first index where "up" occurs

# .isdigit()
vlan = "100"
print(vlan.isdigit())            # True
print("vlan100".isdigit())       # False
```

3.5 String Formatting (f-strings)

> **PyNEng:** "F-strings are the most convenient way of string formatting in Python 3.6+."

```python
```

```python
hostname = "sw-access-01"
ip = "10.10.10.1"
vlan = 100

# f-string (Python 3.6+) — PREFERRED
print(f"Device {hostname} has IP {ip} on VLAN {vlan}")

# Alignment and padding
print(f"{'Hostname':<20} {'IP Address':<16} {'VLAN':>6}")
print(f"{hostname:<20} {ip:<16} {vlan:>6}")

# Debug format (Python 3.8+)
print(f"{hostname=}")            # hostname='sw-access-01'
print(f"{vlan=}")                # vlan=100
```

3.6 Escape Characters

```python
# Newline
print("line1\nline2")

# Tab
print("Name\tIP\tVLAN")
print("sw1\t10.0.0.1\t100")

# Backslash (for Windows paths)
print("C:\\Users\\admin\\scripts")

# Raw strings (ignore escapes) — useful for regex
print(r"C:\Users\admin\scripts")
```

Chapter 3 Lab Tasks

Task 3.1: Given `show_ip = "Interface Gi0/1, IP 192.168.1.1/24, Status up"`, use `.split()` to extract the interface name, IP address (without mask), subnet mask, and status into separate variables. Print each.

Task 3.2: Create a list of 5 Cisco IOS interface configuration commands. Use `"\n".join()` to combine them into a single configuration block string. Print the result.

Task 3.3: Write a script that takes an IPv6 address `"2001:0db8:0000:0042:0000:0000:0000:0001"` and uses string methods to remove leading zeros from each hextet (produce the compressed form).

Task 3.4: Create a formatted table header using f-strings with these columns: Hostname (left-aligned, 20 chars), IP Address (left-aligned, 16 chars), Status (center-aligned, 10 chars). Print 3 sample rows beneath it.

Task 3.5: Given the string `"GigabitEthernet0/0/0"`, write code to check if it starts with "Gig", contains "0/0", and ends with "/0". Print each boolean result.

Chapter 4: Lists

Lists are ordered, mutable collections. In network automation, lists hold device inventories, command sets, interface names, and parsed output lines.

4.1 Creating Lists

```python
# Empty list
my_devices = []

# List with values
vlans = [10, 20, 30, 40, 50]
hostnames = ["rtr1", "rtr2", "sw1", "sw2", "fw1"]
mixed = ["rtr1", 48, True, 3.14]   # mixed types allowed but not recommended

# List from string
show_output = "Gi0/1  10.0.0.1  up\nGi0/2  10.0.0.2  down"
lines = show_output.splitlines()
print(lines)  # ['Gi0/1  10.0.0.1  up', 'Gi0/2  10.0.0.2  down']
```

4.2 Indexing and Slicing

```python
```

```python
devices = ["rtr1", "rtr2", "sw1", "sw2", "fw1"]

# Indexing (0-based)
print(devices[0])      # rtr1 (first)
print(devices[-1])     # fw1 (last)
print(devices[2])      # sw1

# Slicing [start:stop:step]
print(devices[0:2])    # ['rtr1', 'rtr2']  (stop is exclusive)
print(devices[:3])     # ['rtr1', 'rtr2', 'sw1']
print(devices[2:])     # ['sw1', 'sw2', 'fw1']
print(devices[::-1])   # ['fw1', 'sw2', 'sw1', 'rtr2', 'rtr1']  (reversed)
```

4.3 List Methods

KByers Style — Firewall Device Lists

python

```python
#!/usr/bin/env python
from rich import print

locations = ["Munich", "Berlin", "Hamburg", "Frankfurt"]

# Print first and last
print(f"First element: {locations[0]}")
print(f"Last element: {locations[-1]}")
print(f"List length: {len(locations)}")

# .append() — Add one element to the end
locations.append("Leipzig")

# Modify an element by index
locations[3] = "Stuttgart"
print(locations)

# List concatenation
locations = locations + ["Dortmund", "Essen"]
# Alternative: locations += ["Dortmund", "Essen"]
print(locations)

# .pop() — Remove and return an element
city1 = locations.pop(0)     # remove first
city_n = locations.pop()     # remove last (default)
print(f"{city1=}")
print(f"{city_n=}")
print(f"{locations=}")
```

PyNEng Style — More List Methods

```python
```

```python
vlans = [10, 30, 20, 50, 40]

# .sort() — sort in-place
vlans.sort()
print(vlans)                # [10, 20, 30, 40, 50]

# sorted() — returns new sorted list (original unchanged)
original = [50, 20, 40, 10]
new_sorted = sorted(original)
print(original)             # [50, 20, 40, 10]  (unchanged)
print(new_sorted)           # [10, 20, 30, 40, 50]

# .insert(index, value)
vlans.insert(0, 1)
print(vlans)                # [1, 10, 20, 30, 40, 50]

# .remove(value) — removes first occurrence
vlans.remove(30)
print(vlans)                # [1, 10, 20, 40, 50]

# .extend() — add all elements from another list
vlans.extend([60, 70])
print(vlans)                # [1, 10, 20, 40, 50, 60, 70]

# .index(value) — find position
print(vlans.index(40))      # 3

# Check membership
print(20 in vlans)           # True
```

4.4 Iterating Over Lists (Preview — Full coverage in Chapter 8)

```python
firewalls = ["pod1-gaia", "pod2-gaia", "pod3-gaia", "pod4-gaia", "pod5-gaia"]

for fw in firewalls:
    print(fw)

# With index using enumerate
for idx, fw in enumerate(firewalls):
    print(f"{idx} -> {fw}")
```

Chapter 4 Lab Tasks

Task 4.1: Create a list of 8 interface names (e.g., "Gi0/0" through "Gi0/7"). Print the third and last interfaces. Print the total count.

Task 4.2: Given `vlans = [10, 20, 30, 40, 50, 60]`, perform the following in order: (a) append VLAN 70, (b) insert VLAN 5 at position 0, (c) remove VLAN 30, (d) sort the list. Print the result after each step.

Task 4.3: Given two lists `core_devices = ["rtr1", "rtr2"]` and `access_devices = ["sw1", "sw2", "sw3"]`, combine them into a single list called `all_devices` using two different methods. Print both.

Task 4.4: Create a list of 5 IP addresses as strings. Use `.pop()` to remove the first and last. Print the removed IPs and the remaining list.

Task 4.5: Given `config_lines = ["interface Gi0/1", "switchport mode access", "switchport access vlan 10", "no shutdown"]`, reverse the list without using `.reverse()` and print the result.

Chapter 5: Dictionaries

Dictionaries are key-value pairs — the most natural way to represent network device data (hostname → IP, interface → config, etc.).

5.1 Creating Dictionaries

```python
# Empty dictionary
device = {}

# Dictionary with values
device = {
    "hostname": "core-rtr-01",
    "ip": "10.10.10.1",
    "vendor": "Cisco",
    "model": "ISR4431",
    "ios_version": "17.3.4a",
    "interfaces": 48
}
```

KByers Style — Firewall Service Object

```python
#!/usr/bin/env python
from rich import print

ftp_service = {
    "uid": "97aeb3d0-9aea-11d5-bd16-0090272ccb30",
    "name": "ftp",
    "type": "service-tcp",
    "domain": {
        "uid": "a0bbbc99-adef-4ef8-bb6d-defdefdefdef",
        "name": "Check Point Data",
        "domain-type": "data domain",
    },
    "port": "21",
    "icon": "Protocols/FTP",
    "color": "forest green",
}

# Access values
print(ftp_service["name"])            # ftp
print(ftp_service["port"])            # 21

# Nested dictionary access
print(ftp_service["domain"]["name"])  # Check Point Data
```

5.2 Dictionary Operations

KByers Style — Network Object Manipulation

```python
```

```python
#!/usr/bin/env python
from rich import print

network_obj = {
    "name": "hq_net_128",
    "subnet4": "172.31.128.0",
    "mask-length4": 24,
    "color": "green",
    "domain": {"uid": "abc-123-def"}
}

# Retrieve values
name = network_obj["name"]
network = network_obj["subnet4"]
netmask = network_obj["mask-length4"]
print(f"Network Object: {name} ({network}/{netmask})")

# Nested access
domain_uid = network_obj["domain"]["uid"]
print(f"Domain UID: {domain_uid}")

# Add a new key
network_obj["location"] = "Munich"

# Update a value
network_obj["location"] = "Cologne"

# Delete a key
network_obj.pop("color")

print(network_obj)
```

5.3 Dictionary Methods

```python
```

```python
device = {
    "hostname": "sw1",
    "ip": "10.0.0.1",
    "vendor": "Cisco",
}

# .keys(), .values(), .items()
print(list(device.keys()))       # ['hostname', 'ip', 'vendor']
print(list(device.values()))     # ['sw1', '10.0.0.1', 'Cisco']

for key, value in device.items():
    print(f"{key}: {value}")

# .get() — safe access (returns None if key missing, no error)
model = device.get("model")
print(model)                     # None
model = device.get("model", "Unknown")
print(model)                     # Unknown

# .update() — merge dictionaries
extra_info = {"model": "C9300", "ports": 48}
device.update(extra_info)
print(device)

# .setdefault() — set value only if key does NOT exist
device.setdefault("vendor", "Arista")   # no change, "Cisco" already exists
device.setdefault("site", "HQ")         # adds "site": "HQ"

# Check membership (checks KEYS by default)
print("hostname" in device)     # True
print("sw1" in device)          # False (that's a value, not a key)
```

5.4 Netmiko Device Dictionaries

> **Netmiko Course:** Device parameters are always passed as dictionaries.

```python
```

```python
from netmiko import ConnectHandler

# Standard Netmiko device dictionary
cisco_device = {
    "device_type": "cisco_ios",
    "host": "cisco3.lasthop.io",
    "username": "pyclass",
    "password": "secret123",
}

# Dictionary unpacking with **
net_connect = ConnectHandler(**cisco_device)
# This is equivalent to:
# ConnectHandler(device_type="cisco_ios", host="cisco3.lasthop.io", ...)
```

Chapter 5 Lab Tasks

Task 5.1: Create a dictionary for a network device with keys: hostname, ip, vendor, model, os_version, site. Access and print each value using both bracket notation and `.get()`.

Task 5.2: Given a nested dictionary representing a firewall rule, access the "destination" IP which is nested 2 levels deep. Print it.

```python
fw_rule = {
    "name": "Allow_Web",
    "source": {"type": "host", "ip": "10.0.0.5"},
    "destination": {"type": "network", "ip": "192.168.1.0", "mask": "/24"},
    "action": "permit"
}
```

Task 5.3: Create two dictionaries for two routers. Merge them into a single `inventory` dictionary where the keys are hostnames and values are the full device dictionaries.

Task 5.4: Loop through a device dictionary using `.items()` and print each key-value pair formatted as `"key: value"` in a nicely aligned table.

Task 5.5: Create a Netmiko-style device dictionary for an Arista switch. Add a `session_log` key after creation. Change the `device_type` from `"arista_eos"` to `"arista_eos_ssh"`. Delete the `session_log` key. Print the final dictionary.

Chapter 6: Tuples, Sets, and Booleans

6.1 Tuples

Tuples are immutable lists. They are used when data should not change (e.g., a fixed set of SNMP community strings, or coordinates).

```python
# Create tuples
dns_servers = ("8.8.8.8", "8.8.4.4")
credentials = ("admin", "secret123")

# Indexing (same as lists)
print(dns_servers[0])           # 8.8.8.8

# Unpacking
primary, secondary = dns_servers
print(primary)                  # 8.8.8.8

# Tuples are IMMUTABLE
# dns_servers[0] = "1.1.1.1"   # TypeError!

# Single element tuple requires trailing comma
single = ("only_one",)
not_a_tuple = ("only_one")     # This is just a string!
```

6.2 Sets

Sets are unordered collections of unique elements. They are excellent for comparing configurations, finding differences in VLAN lists, etc.

```python
```

```python
# Create sets
vlans_sw1 = {10, 20, 30, 40, 50}
vlans_sw2 = {30, 40, 50, 60, 70}

# Union — all VLANs across both switches
all_vlans = vlans_sw1 | vlans_sw2
print(all_vlans)                # {10, 20, 30, 40, 50, 60, 70}

# Intersection — VLANs on BOTH switches
common = vlans_sw1 & vlans_sw2
print(common)                   # {30, 40, 50}

# Difference — VLANs on sw1 but NOT sw2
only_sw1 = vlans_sw1 - vlans_sw2
print(only_sw1)                 # {10, 20}

# Symmetric difference — VLANs on one OR the other but NOT both
unique = vlans_sw1 ^ vlans_sw2
print(unique)                   # {10, 20, 60, 70}

# Practical: Find blocked IPs to add/remove
current_blocked = {"1.2.3.4", "5.6.7.8", "9.10.11.12"}
new_blocked = {"5.6.7.8", "9.10.11.12", "13.14.15.16"}

add_ips = new_blocked - current_blocked
remove_ips = current_blocked - new_blocked
print(f"Add: {add_ips}")        # Add: {'13.14.15.16'}
print(f"Remove: {remove_ips}")  # Remove: {'1.2.3.4'}
```

KByers Style — Set Comparison for Blocked IPs

```python
# From class4/exercises/main_project/03_mgmt_api_cfg.py
current_blocked_ips = get_current_blocked_ips(api_client, group_name)
new_blocked_ips = read_blocked_ips_file()

if set(current_blocked_ips) == set(new_blocked_ips):
    print("No Blocked IP Changes")
else:
    add_blocked_ips = set(new_blocked_ips) - set(current_blocked_ips)
    remove_blocked_ips = set(current_blocked_ips) - set(new_blocked_ips)
```

6.3 Booleans and Truthiness

KByers Style — Truthiness in Python

```python
#!/usr/bin/env python
from rich import print

# Strings: empty string is False
some_str = ""
print(bool(some_str))           # False
print(bool("hello"))            # True

# Numbers: 0 is False
some_int = 0
print(bool(some_int))           # False
print(bool(42))                 # True

# Lists: empty list is False
some_list = []
print(bool(some_list))          # False
print(bool([1, 2]))             # True

# Dict: empty dict is False
some_dict = {}
print(bool(some_dict))          # False
```

Short-Circuit Evaluation

```python
# and: if first condition is False, skip second
check_cond = False
if check_cond and some_function():    # some_function() never called
    print("Never printed")

# or: if first condition is True, skip second
skip_func = True
if skip_func or some_function():      # some_function() never called
    print("Always printed")
```

Chapter 6 Lab Tasks

Task 6.1: Create a tuple of 3 DNS server IPs. Try to modify the first element. What error do you get? Unpack all three into individual variables and print them.

Task 6.2: Two switches have these VLANs: `sw1 = {10, 20, 30, 40, 50}` and `sw2 = {30, 40, 50, 60, 70}`. Find: (a) all VLANs, (b) common VLANs, (c) VLANs only on sw1, (d) VLANs unique to one switch.

Task 6.3: Create a list with duplicates: `[10, 20, 30, 10, 20, 40]`. Convert it to a set to remove duplicates, then back to a sorted list. Print each step.

Task 6.4: Write code that checks the "truthiness" of: `0`, `""`, `[]`, `{}`, `None`, `"hello"`, `[1]`, `42`. Print the boolean value of each.

Task 6.5: Given `current_acl = {"permit_web", "permit_ssh", "deny_telnet"}` and `desired_acl = {"permit_web", "permit_ssh", "permit_dns"}`, find which rules to add and which to remove. Print both.

Chapter 7: Conditionals (if/elif/else)

7.1 Basic if/elif/else

> **PyNEng:** "The if/elif/else statement allows make branches during program execution."

LPTHW Style — Simple Decisions

```python
people = 20
cats = 30
dogs = 15

if people < cats:
    print("Too many cats! The world is doomed!")
elif people > cats:
    print("Not many cats! The world is saved!")
else:
    print("The world is on a thin line.")
```

KByers Style — Firewall Pod Selection

```python
```

```python
#!/usr/bin/env python
from rich import print

pod_numb = input("Enter pod number: ")
fw_name = f"chkpnt-pod{pod_numb}"

if fw_name == "chkpnt-pod1":
    print("Found pod1")
elif fw_name == "chkpnt-pod99":
    print("Found pod99")
else:
    print("Not my pod")
```

7.2 Compound Conditions (and, or, not)

KByers Style — API Version Checking

```python
#!/usr/bin/env python
from rich import print

api_version = "1.8"

if "1." in api_version:
    base_api = "v1"
elif "2." in api_version:
    base_api = "v2"

# Compound conditions
if base_api == "v1" and float(api_version) >= 1.8:
    print("API Version 1.8 or later (not V2)")

if base_api == "v1" or base_api == "v2":
    print("API V1 or V2")
```

7.3 Checking Data Structures

PyNEng Style

```python
```

```python
# Empty list check (Pythonic way)
route_table = []
if route_table:
    print("Routes found")
else:
    print("No routes in table")

# Membership check
allowed_vlans = [10, 20, 30, 40, 50]
new_vlan = 25
if new_vlan in allowed_vlans:
    print(f"VLAN {new_vlan} is allowed")
else:
    print(f"VLAN {new_vlan} is NOT allowed")
```

7.4 Nested Conditionals

KByers Style — Service Matching with JSON

```python
#!/usr/bin/env python
import json
from rich import print

with open("tcp_services.json") as f:
    services = json.load(f)

service_list = services["objects"]
print(len(service_list))

for service in service_list:
    service_name = service["name"]
    if service_name == "ftp":
        print("Found FTP Service")
    elif service_name == "domain-tcp":
        print("Found DNS (TCP)")
    else:
        print(service_name)
```

7.5 Ternary Expressions

```python
# Compact conditional assignment
interface_status = "up"
color = "green" if interface_status == "up" else "red"
print(f"Status: [{color}]{interface_status}")

# From Netmiko course — password handling
import os
from getpass import getpass
password = os.getenv("NETMIKO_PASSWORD") if os.getenv("NETMIKO_PASSWORD") else getpa
```

Chapter 7 Lab Tasks

Task 7.1: Write a script that takes a VLAN number as input. If the VLAN is 1, print "Default VLAN — do not use". If the VLAN is between 2-1000, print "Standard VLAN". If between 1001-4094, print "Extended VLAN". Otherwise, print "Invalid VLAN".

Task 7.2: Given an interface status dictionary `{"name": "Gi0/1", "status": "up", "protocol": "up", "speed": "1000"}`, write code that checks: if both status AND protocol are "up", print "Fully operational". If status is "up" but protocol is "down", print "Layer 1 up, Layer 2 issue". Otherwise print "Interface down".

Task 7.3: Write a script that checks if a given IP address string is valid (simple check: split by ".", check exactly 4 octets, each octet is a digit between 0-255). Print "Valid" or "Invalid".

Task 7.4: Given `api_version = "1.8"`, use conditionals to determine the base URL. If version starts with "1.", use `base_api = "v1"`. If version starts with "2.", use `base_api = "v2"`. Print the resulting `base_url` as `https://host/{base_api}/api/`.

Task 7.5: Write a ternary expression that assigns "SSH" if port is 22, "HTTPS" if port is 443, or "Unknown" for any other port. Test with ports 22, 443, and 8080.

Chapter 8: Loops (for and while)

8.1 for Loops

KByers Style — Iterating Over Firewalls

```python
from rich import print

my_firewalls = [
    "pod1-gaia",
    "pod2-gaia",
    "pod3-gaia",
    "pod4-gaia",
    "pod5-gaia",
]

# Basic for loop
for fw in my_firewalls:
    print(fw)

# Conditional inside loop
for fw in my_firewalls:
    if fw == "pod5-gaia":
        print(f"\nConnecting to fw: {fw}\n")
```

PyNEng Style — Generating Config

```python
```

```
access_template = [
    'switchport mode access',
    'switchport access vlan {}',
    'switchport nonegotiate',
    'spanning-tree portfast',
    'spanning-tree bpduguard enable'
]

vlans = [10, 20, 30]
for vlan in vlans:
    print(f"interface Vlan{vlan}")
    for command in access_template:
        print(f" {command.format(vlan)}")
    print("!")
```

8.2 break and continue

KByers Style

```python
from rich import print

my_firewalls = [
    "pod1-gaia", "pod2-gaia", "pod3-gaia",
    "pod4-gaia", "pod5-gaia", "pod98-gaia", "pod99-gaia",
]

for fw in my_firewalls:
    if fw == "pod4-gaia":
        continue          # Skip pod4, go to next iteration
    if fw == "pod98-gaia":
        break             # Stop the entire loop
    print(fw)

# Output: pod1-gaia, pod2-gaia, pod3-gaia, pod5-gaia
```

8.3 enumerate() — Loop with Index

```python
```

```python
from rich import print

my_firewalls = ["pod1-gaia", "pod2-gaia", "pod3-gaia"]

for idx, fw in enumerate(my_firewalls):
    print(f"{idx} -> {fw}")

# Start counting from 1
for idx, fw in enumerate(my_firewalls, start=1):
    print(f"Device #{idx}: {fw}")
```

8.4 Looping Over Dictionaries

```python
device = {
    "hostname": "rtr1",
    "ip": "10.0.0.1",
    "vendor": "Cisco",
    "model": "ISR4431"
}

# Loop over keys
for key in device:
    print(key)

# Loop over values
for value in device.values():
    print(value)

# Loop over key-value pairs (most common)
for key, value in device.items():
    print(f"{key:>12}: {value}")
```

8.5 while Loops

KByers Style — Wait with Timeout

```python
```

```python
from rich import print
import time

WAIT_TIME = 8

start_time = time.time()
while True:
    cur_time = time.time()
    if cur_time - start_time < WAIT_TIME:
        print("Sleeping 1s")
        time.sleep(1)
    else:
        print("All done...")
        break
```

PyNEng Style — User Input Loop

```python
while True:
    password = input("Enter password (min 8 chars): ")
    if len(password) >= 8:
        print("Password accepted")
        break
    print("Too short, try again")
```

8.6 Nested Loops

```python
# Generate interface configs for multiple switches
switches = ["sw1", "sw2"]
interfaces = ["Gi0/1", "Gi0/2", "Gi0/3"]
vlan = 10

for switch in switches:
    print(f"\n! Configuration for {switch}")
    for intf in interfaces:
        print(f"interface {intf}")
        print(f" switchport access vlan {vlan}")
        vlan += 10
```

Chapter 8 Lab Tasks

Task 8.1: Create a list of 10 IP addresses (e.g., "10.0.0.1" through "10.0.0.10"). Use a for loop to print each IP with its index number starting from 1.

Task 8.2: Loop through a list of device hostnames. Use `continue` to skip any hostname containing "test". Use `break` to stop when you encounter "STOP". Print all other hostnames.

```python
devices = ["rtr1", "sw1", "test-rtr", "fw1", "STOP", "rtr2"]
```

Task 8.3: Write a while loop that simulates a retry mechanism: attempt to "connect" (just print a message) up to 3 times, with a 1-second sleep between attempts. After 3 failures, print "Connection failed after 3 retries".

Task 8.4: Given a dictionary of devices, loop through it and print a formatted table:

```python
devices = {
    "rtr1": {"ip": "10.0.0.1", "vendor": "Cisco"},
    "sw1": {"ip": "10.0.0.2", "vendor": "Arista"},
    "fw1": {"ip": "10.0.0.3", "vendor": "Palo Alto"},
}
```

Task 8.5: Write nested loops that generate trunk interface configs for 3 interfaces across 2 switches, with allowed VLANs 10, 20, 30. The output should look like real Cisco IOS configuration.

Chapter 9: Functions

9.1 Defining Functions

LPTHW Style — Building Blocks

```python
```

```python
def print_two(*args):
    arg1, arg2 = args
    print(f"arg1: {arg1}, arg2: {arg2}")

def print_one(arg1):
    print(f"arg1: {arg1}")

def print_none():
    print("I got nothing.")

print_two("Zed", "Shaw")
print_one("First!")
print_none()
```

KByers Style — Network Functions

```python
#!/usr/bin/env python
from rich import print

def fw_func(name, ipaddr, os_version="R82"):
    print(f"{name=}")
    print(f"{ipaddr=}")
    print(f"{os_version=}")
    return f"{name}-{ipaddr}-{os_version}"

# Positional arguments
ret_val = fw_func("chkpnt-pod99", "3.77.44.109", "R81.20")
print(f"\n{ret_val=}\n")

# Named (keyword) arguments
ret_val = fw_func(
    os_version="R82",
    ipaddr="3.77.44.100",
    name="chkpnt-pod1",
)
print(f"\n{ret_val=}\n")

# Using default value
ret_val = fw_func(name="chkpnt-pod1", ipaddr="3.77.44.100")
print(f"\n{ret_val=}\n")
```

9.2 Return Values

```python
def check_interface_status(status, protocol):
    """Check if interface is fully operational."""
    if status == "up" and protocol == "up":
        return "operational"
    elif status == "up":
        return "layer2_issue"
    else:
        return "down"

result = check_interface_status("up", "up")
print(result)    # operational

# Functions without return statement return None
def greet(name):
    print(f"Hello {name}")

result = greet("Engineer")    # prints "Hello Engineer"
print(result)                  # None
```

9.3 Docstrings

```python
```

```python
def calculate_subnet_hosts(prefix_length):
    """Calculate the number of usable hosts in a subnet.

    Args:
        prefix_length (int): CIDR prefix length (0-32)

    Returns:
        int: Number of usable host addresses

    Example:
        >>> calculate_subnet_hosts(24)
        254
    """
    if prefix_length < 0 or prefix_length > 32:
        raise ValueError(f"Invalid prefix: {prefix_length}")
    total = 2 ** (32 - prefix_length)
    return total - 2 if prefix_length < 31 else total
```

9.4 Functions as Reusable Modules

KByers Style — API Helper Functions

```python
```

```python
import requests
import json

def login(base_url, user, password, ssl_verify=False):
    """Login and return session headers."""
    url = base_url + "login"
    headers = {"Content-Type": "application/json"}
    login_payload = {"user": user, "password": password}
    response = requests.post(
        url, data=json.dumps(login_payload),
        headers=headers, verify=ssl_verify
    )
    return response

def api_call(base_url, endpoint, headers, payload=None, ssl_verify=False):
    """Make an API call with session headers."""
    if payload is None:
        payload = {}
    url = base_url + endpoint
    response = requests.post(
        url, data=json.dumps(payload),
        headers=headers, verify=ssl_verify
    )
    return response

def logout(base_url, headers, ssl_verify=False):
    """Logout and clean up session."""
    endpoint = "logout"
    response = api_call(base_url, endpoint, headers, ssl_verify=ssl_verify)
    if response.status_code == 200:
        if "X-chkp-sid" in headers:
            headers.pop("X-chkp-sid")
    return response
```

9.5 *args and **kwargs

```python
```

```python
# *args — variable number of positional arguments
def configure_vlans(*vlans):
    for vlan in vlans:
        print(f" vlan {vlan}")

configure_vlans(10, 20, 30)

# **kwargs — variable number of keyword arguments
def create_device(**params):
    for key, value in params.items():
        print(f"  {key}: {value}")

create_device(hostname="rtr1", ip="10.0.0.1", vendor="Cisco")

# Dictionary unpacking with **
device = {"device_type": "cisco_ios", "host": "10.0.0.1", "username": "admin"}
# ConnectHandler(**device) unpacks the dict into keyword arguments
```

Chapter 9 Lab Tasks

Task 9.1: Write a function `generate_hostname(site, role, number)` that returns a hostname like "NYC-RTR-01". Include a default value of "01" for number. Test with several calls.

Task 9.2: Write a function `validate_ip(ip_address)` that takes an IP address string and returns `True` if valid, `False` otherwise. Include a docstring explaining the function.

Task 9.3: Write two functions: `read_json(filename)` that reads a JSON file and returns the data structure, and `extract_fields(task)` that extracts specific fields from a dict. Call one from the other.

Task 9.4: Write a function that accepts `**kwargs` for device parameters (hostname, ip, vendor, model) and prints a formatted device summary. Handle missing keys with `.get()` defaults.

Task 9.5: Refactor the following code into a reusable function that can work for any device list:

```python
for device in devices:
    print(f"Connecting to {device['hostname']}...")
    print(f"  IP: {device['ip']}")
    print(f"  Vendor: {device['vendor']}")
```

Chapter 10: Working with Files

10.1 Reading Files

KByers Style — Multiple Methods

```python
#!/usr/bin/env python
from rich import print

# Method 1: read() — entire file as one string
f = open("my_file.txt", "r")
data = f.read()
print(data)
f.close()

# Method 2: readlines() — file as list of lines
f = open("my_file.txt", "r")
lines = f.readlines()
print(type(lines))     # <class 'list'>
print(lines)           # ['line1\n', 'line2\n', ...]
f.close()

# Method 3: iterate line by line
f = open("my_file.txt", "r")
for line in f:
    line = line.strip()    # remove trailing newline
    print(repr(line))
f.close()
```

10.2 Context Manager (with statement)

> **PyNEng & KByers:** Always use `with` to automatically close files.

```python
# Context manager automatically closes the file when done
with open("my_file.txt") as f:
    data = f.read()

print(data)
# File is already closed here — no need for f.close()
```

10.3 Writing Files

KByers Style

```python
#!/usr/bin/env python

# Write (creates or overwrites)
with open("new_file.txt", "w") as f:
    f.write("--> hello world <--\n")
    f.write("something else\n")
    f.write("one last line\n")

# Append (adds to end of existing file)
with open("new_file.txt", "a") as f:
    f.write("appended line\n")
```

10.4 Reading JSON Files

```python
import json
from rich import print

# Read JSON
with open("gaia_api.json") as f:
    data = json.load(f)

print(type(data))    # <class 'dict'>
print(data)

# Write JSON
data = {
    "current-version": "1.8",
    "supported-versions": ["1", "1.1", "1.2", "1.3", "1.4", "1.5", "1.6", "1.7", "1.
}

with open("gaia_api.json", "w") as f:
    json.dump(data, f, indent=4)
```

10.5 Reading YAML Files

```python
import yaml
from rich import print

# Read YAML
with open("gaia_api.yaml") as f:
    data = yaml.safe_load(f)

print(data)

# Write YAML
data = {
    "current-version": "1.8",
    "supported-versions": ["1", "1.1", "1.2", "1.3", "1.4", "1.5", "1.6", "1.7", "1.
}

with open("gaia_api.yaml", "w") as f:
    yaml.dump(data, f, default_flow_style=False)
```

10.6 Read vs Load: String vs Data Structure

KByers Style — Critical Distinction

```python
import json
from rich import print

filename = "network_objects.json"

# Read as TEXT (string)
with open(filename) as f:
    data = f.read()
print(f"Type: {type(data)}")       # <class 'str'>

# Read as DATA STRUCTURE (dict/list)
with open(filename) as f:
    data = json.load(f)
print(f"Type: {type(data)}")       # <class 'dict'>
print(f"Type of 'objects': {type(data['objects'])}")  # <class 'list'>
```

Chapter 10 Lab Tasks

Task 10.1: Create a text file with 5 router hostnames (one per line). Read the file and print each hostname stripped of whitespace.

Task 10.2: Write a script that reads a JSON file containing a list of devices, loops through them, and prints each device's hostname and IP. Create the JSON file first.

Task 10.3: Create a YAML file with a list of firewall names and their management IPs. Read it back and print the data.

Task 10.4: Write a script that reads a config file line by line, counts the number of "interface" lines, and writes only those interface lines to a new file.

Task 10.5: Demonstrate the difference between `f.read()` (string), `json.load()` (data structure), and `f.readlines()` (list of strings) on the same JSON file. Print the type of each.

Chapter 11: Working with CSV, JSON, and YAML

11.1 CSV Files

```python
```

```python
import csv

# Read CSV
with open("devices.csv") as f:
    reader = csv.DictReader(f)
    for row in reader:
        print(f"{row['hostname']:15} {row['ip']:16} {row['vendor']}")

# Write CSV
devices = [
    {"hostname": "rtr1", "ip": "10.0.0.1", "vendor": "Cisco"},
    {"hostname": "sw1", "ip": "10.0.0.2", "vendor": "Arista"},
]

with open("devices.csv", "w", newline="") as f:
    fieldnames = ["hostname", "ip", "vendor"]
    writer = csv.DictWriter(f, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(devices)
```

11.2 JSON — Deep Dive

```python
```

```python
import json

# json.loads() — parse JSON STRING into Python object
json_string = '{"hostname": "rtr1", "ip": "10.0.0.1"}'
data = json.loads(json_string)
print(type(data))    # <class 'dict'>

# json.dumps() — convert Python object to JSON STRING
python_dict = {"hostname": "rtr1", "interfaces": ["Gi0/1", "Gi0/2"]}
json_str = json.dumps(python_dict, indent=2)
print(json_str)

# json.load() — read JSON from FILE
with open("data.json") as f:
    data = json.load(f)

# json.dump() — write Python object to JSON FILE
with open("output.json", "w") as f:
    json.dump(data, f, indent=4)
```

11.3 YAML — Deep Dive

yaml

```yaml
# devices.yaml
---
- hostname: rtr1
  ip: 10.0.0.1
  vendor: Cisco
  interfaces:
    - Gi0/1
    - Gi0/2
- hostname: sw1
  ip: 10.0.0.2
  vendor: Arista
```

python

```python
import yaml

with open("devices.yaml") as f:
    devices = yaml.safe_load(f)

for device in devices:
    print(f"{device['hostname']}: {device['ip']}")
```

Chapter 11 Lab Tasks

Task 11.1: Create a CSV file with 5 devices (hostname, ip, vendor, model). Read it with `csv.DictReader` and print a formatted table.

Task 11.2: Convert this Python dictionary to a JSON string with indentation, then parse it back. Verify the round-trip.

```python
network = {"name": "HQ_LAN", "subnet": "10.0.0.0/8", "vlans": [10, 20, 30]}
```

Task 11.3: Create a YAML file representing 3 devices with nested interface lists. Read it and print each device's hostname and interface count.

Task 11.4: Write a script that reads a JSON file, adds a new field to each device object, and writes the modified data to a new JSON file.

Task 11.5: Convert a CSV device inventory to YAML format programmatically (read CSV, write YAML).

Chapter 12: Error Handling (try/except)

12.1 Basic try/except

KByers Style — Progressive Error Handling

```python
```

```python
#!/usr/bin/env python
from rich import print

# Basic try/except
try:
    result = 10 / 0
except ZeroDivisionError:
    print("Cannot divide by zero!")

# Multiple exception types
try:
    data = {"key": "value"}
    print(data["missing_key"])
except KeyError as e:
    print(f"Key not found: {e}")
except TypeError as e:
    print(f"Type error: {e}")

# Generic exception (catch-all)
try:
    risky_operation()
except Exception as e:
    print(f"Something went wrong: {e}")

# try/except/else/finally
try:
    with open("config.json") as f:
        data = json.load(f)
except FileNotFoundError:
    print("Config file not found!")
    data = {}
except json.JSONDecodeError:
    print("Invalid JSON!")
    data = {}
else:
    print("Config loaded successfully")
finally:
    print("Cleanup complete")
```

12.2 Custom Exceptions

KByers Style — Security Automation Exceptions

```python
class ChkPntConfigError(Exception):
    """Raised when Check Point configuration fails."""
    pass

class ChkPntPolicyInstallError(Exception):
    """Raised when firewall policy installation fails."""
    pass

class ChkptAuthError(Exception):
    """Raised when session ID is missing or expired."""
    pass

# Using custom exceptions
def api_call(url, headers, payload=None):
    if "X-chkp-sid" not in headers:
        msg = "Session ID not set, please call '.login()' first."
        raise ChkptAuthError(msg)
    # ... rest of implementation
```

Chapter 12 Lab Tasks

Task 12.1: Write a script that tries to open a file that doesn't exist. Catch the `FileNotFoundError` and print a friendly message.

Task 12.2: Write a function that converts a string to an integer. Handle `ValueError` for non-numeric strings. Return `None` on failure.

Task 12.3: Create a custom exception `InvalidVlanError`. Write a function that raises it when a VLAN number is outside 1-4094.

Task 12.4: Write a try/except/else/finally block that reads a JSON file: handle missing file and invalid JSON separately, print "success" in `else`, and "done" in `finally`.

Task 12.5: Write a function that connects to a device (simulate with a dict lookup). If the device is not in the inventory, raise a custom `DeviceNotFoundError` with the hostname in the message.

Chapter 13: Modules and Packages

13.1 Importing Modules

```python
# Import entire module
import os
print(os.getcwd())

# Import specific function
from os import path
print(path.exists("/etc/hosts"))

# Import with alias
import json as j
data = j.loads('{"key": "value"}')

# Import multiple items
from pathlib import Path
from rich import print
```

13.2 Creating Your Own Module

```python
# File: network_utils.py
def validate_ip(ip_address):
    octets = ip_address.split(".")
    if len(octets) != 4:
        return False
    return all(o.isdigit() and 0 <= int(o) <= 255 for o in octets)

def generate_hostname(site, role, number=1):
    return f"{site}-{role}-{number:02d}"
```

```python
# File: main.py
from network_utils import validate_ip, generate_hostname

print(validate_ip("10.0.0.1"))           # True
print(generate_hostname("NYC", "RTR"))    # NYC-RTR-01
```

13.3 The `if __name__ == "__main__"` Pattern

```python
# File: gaia_funcs.py
def login(url, username, password):
    """Login and return session ID."""
    # ... implementation

def logout(url, headers):
    """Logout from API."""
    # ... implementation

if __name__ == "__main__":
    # This code ONLY runs when script is executed directly
    # NOT when imported as a module
    print("Testing functions...")
    session_id = login("https://fw.example.com/api/", "admin", "pass")
    print(f"Got session: {session_id}")
```

Chapter 13 Lab Tasks

Task 13.1: Create a module called `net_helpers.py` with 3 functions: `validate_ip()`, `mask_to_cidr()`, and `generate_hostname()`. Import and use them from a separate script.

Task 13.2: Demonstrate 3 different import styles (import, from...import, import...as) using the `json` module.

Task 13.3: Add `if __name__ == "__main__"` to your `net_helpers.py` module that runs test cases when executed directly.

Task 13.4: Create a package (folder with `__init__.py`) called `netauto` containing two modules: `devices.py` and `configs.py`. Import from both in a main script.

Task 13.5: Install the `rich` library and use `from rich import print` alongside `from rich.table import Table` to create a formatted device table.

Chapter 14: Useful Built-in Functions

14.1 range()

```python
# Generate VLAN IDs
for vlan in range(10, 51, 10):   # start, stop, step
    print(f"vlan {vlan}")
# Output: vlan 10, vlan 20, vlan 30, vlan 40, vlan 50
```

14.2 enumerate()

```python
interfaces = ["Gi0/1", "Gi0/2", "Gi0/3"]
for idx, intf in enumerate(interfaces, start=1):
    print(f"{idx}. {intf}")
```

14.3 zip()

```python
hostnames = ["rtr1", "rtr2", "rtr3"]
ips = ["10.0.0.1", "10.0.0.2", "10.0.0.3"]
vendors = ["Cisco", "Cisco", "Juniper"]

for hostname, ip, vendor in zip(hostnames, ips, vendors):
    print(f"{hostname:10} {ip:15} {vendor}")
```

14.4 sorted(), map(), filter(), lambda

```python
```

```python
# sorted() with key
devices = [
    {"name": "sw2", "ip": "10.0.0.2"},
    {"name": "rtr1", "ip": "10.0.0.1"},
    {"name": "fw1", "ip": "10.0.0.3"},
]
sorted_devices = sorted(devices, key=lambda d: d["name"])

# map() — apply function to every element
vlans = ["10", "20", "30"]
vlan_ints = list(map(int, vlans))    # [10, 20, 30]

# filter() — keep elements that pass test
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
evens = list(filter(lambda x: x % 2 == 0, numbers))  # [2, 4, 6, 8, 10]

# all() / any()
statuses = ["up", "up", "up"]
print(all(s == "up" for s in statuses))    # True

statuses = ["up", "down", "up"]
print(any(s == "down" for s in statuses))  # True
```

Chapter 14 Lab Tasks

Task 14.1: Use `range()` to generate a list of IPs from 10.0.0.1 to 10.0.0.10. Print each.

Task 14.2: Use `zip()` to combine three lists (hostnames, IPs, VLANs) into a list of dictionaries.

Task 14.3: Use `sorted()` with a `lambda` key to sort a list of device dictionaries by IP address.

Task 14.4: Use `filter()` to extract only "up" interfaces from a list of interface status dicts.

Task 14.5: Use `all()` to verify every device in a list has the key "hostname" set to a non-empty string.

Chapter 15: Regular Expressions

15.1 Basic Regex with re Module

```python
```

```python
import re

# re.search() — find first match
line = "Interface GigabitEthernet0/0/0 is up, line protocol is up"
match = re.search(r"(\S+) is (up|down)", line)
if match:
    print(match.group(0))    # GigabitEthernet0/0/0 is up
    print(match.group(1))    # GigabitEthernet0/0/0
    print(match.group(2))    # up

# re.findall() — find ALL matches
config = """
ip route 10.0.0.0 255.0.0.0 192.168.1.1
ip route 172.16.0.0 255.240.0.0 192.168.1.1
ip route 0.0.0.0 0.0.0.0 192.168.1.254
"""
routes = re.findall(r"ip route (\S+) (\S+) (\S+)", config)
for dest, mask, nexthop in routes:
    print(f"Destination: {dest}/{mask} via {nexthop}")
```

15.2 Common Regex Patterns for Network Engineers

```python
```

```python
import re

# IP address pattern
ip_pattern = r"\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}"

# MAC address pattern
mac_pattern = r"[0-9a-fA-F]{4}\.[0-9a-fA-F]{4}\.[0-9a-fA-F]{4}"

# Interface name pattern
intf_pattern = r"(Gi|Te|Fa|Lo|Vl)\S+"

# VLAN pattern
vlan_pattern = r"Vlan(\d+)"

text = "Interface Gi0/1 has IP 10.0.0.1 and MAC aabb.ccdd.eeff on Vlan100"
print(re.findall(ip_pattern, text))      # ['10.0.0.1']
print(re.findall(mac_pattern, text))     # ['aabb.ccdd.eeff']
print(re.findall(intf_pattern, text))    # ['Gi0/1']
print(re.findall(vlan_pattern, text))    # ['100']
```

15.3 re.sub() — Find and Replace

```python
import re

config = "interface GigabitEthernet0/1\n ip address 10.0.0.1 255.255.255.0"
new_config = re.sub(r"10\.0\.0\.1", "192.168.1.1", config)
print(new_config)
```

Chapter 15 Lab Tasks

Task 15.1: Use `re.search()` to extract the IP address and subnet mask from: `"ip address 192.168.1.1 255.255.255.0"`.

Task 15.2: Use `re.findall()` to extract all IP addresses from a multi-line `show ip route` output string.

Task 15.3: Write a regex that matches Cisco interface names (GigabitEthernet, TenGigabitEthernet, FastEthernet, Loopback, Vlan). Test against a sample config.

Task 15.4: Use `re.sub()` to replace all occurrences of an old IP with a new IP in a config string.

Task 15.5: Parse the following line to extract hostname, IP, and status using named groups: `"Device: rtr1, IP: 10.0.0.1, Status: up"`

Chapter 16: List Comprehensions and Generators

16.1 List Comprehensions

```python
# Traditional loop
vlans = []
for i in range(10, 60, 10):
    vlans.append(i)

# List comprehension (same result, one line)
vlans = [i for i in range(10, 60, 10)]
print(vlans)    # [10, 20, 30, 40, 50]

# With condition
up_interfaces = [intf for intf in interfaces if intf["status"] == "up"]

# Transform + filter
hostnames = ["RTR1", "SW1", "FW1"]
lower_names = [name.lower() for name in hostnames]

# Network examples
ips = [f"10.0.0.{i}" for i in range(1, 11)]
print(ips)  # ['10.0.0.1', '10.0.0.2', ..., '10.0.0.10']

# From KByers — strip whitespace from file lines
with open("blocked_ips.txt") as f:
    blocked_ips = [ip.strip() for ip in f.readlines()]

# From KByers — generate host objects
blocked_ip_objs = [gen_host_object(ip) for ip in add_blocked_ips]
```

16.2 Dictionary Comprehensions

```python
```

```python
# Create dict from two lists
hostnames = ["rtr1", "rtr2", "rtr3"]
ips = ["10.0.0.1", "10.0.0.2", "10.0.0.3"]

device_map = {host: ip for host, ip in zip(hostnames, ips)}
print(device_map)  # {'rtr1': '10.0.0.1', 'rtr2': '10.0.0.2', 'rtr3': '10.0.0.3'}
```

16.3 Generator Expressions

```python
# Generator — lazy evaluation, saves memory for large datasets
ip_gen = (f"10.0.0.{i}" for i in range(1, 1000001))

# Only generates values as needed
for ip in ip_gen:
    if ip == "10.0.0.5":
        print(f"Found: {ip}")
        break
```

Chapter 16 Lab Tasks

Task 16.1: Use a list comprehension to create a list of 20 IPs from "192.168.1.1" to "192.168.1.20".

Task 16.2: Given a list of interface dicts, use a list comprehension to extract only interfaces where status is "up".

Task 16.3: Use a dictionary comprehension to invert a dictionary (swap keys and values).

Task 16.4: Read a file of IP addresses and use a list comprehension to strip whitespace and filter out comment lines (starting with #).

Task 16.5: Rewrite this loop as a list comprehension:

```python
results = []
for device in devices:
    if device["vendor"] == "Cisco":
        results.append(device["hostname"].upper())
```

Chapter 17: Object-Oriented Programming — Basics

17.1 Why OOP for Network Automation?

Classes let you model network objects: a Router, a Firewall, an API Client. Each has attributes (hostname, IP) and methods (connect, send_command, logout).

17.2 Creating a Class

KByers Style — Check Point API Class

```python
class ChkptAPI:
    def __init__(self, host, username, password, mode="web_api",
                 api_version=None, ssl_verify=False):
        self.host = host
        self.username = username
        self.password = password

        if api_version is None:
            if mode == "web_api":
                api_version = "2"
            elif mode == "gaia_api":
                api_version = "1.8"

        self.ssl_verify = ssl_verify
        self.headers = {"Content-Type": "application/json"}
        self.base_url = f"https://{host}/{mode}/v{api_version}/"

# Usage
api_client = ChkptAPI(
    host="chkpnt-pod99.lasthop.io",
    username="admin",
    password="testpass",
    mode="gaia_api"
)
print(api_client.base_url)
# https://chkpnt-pod99.lasthop.io/gaia_api/v1.8/
```

17.3 Methods

```python
```

```python
import requests
import json

class ChkptAPI:
    def __init__(self, host, username, password, mode="web_api",
                 api_version=None, ssl_verify=False):
        self.host = host
        self.username = username
        self.password = password
        if api_version is None:
            api_version = "2" if mode == "web_api" else "1.8"
        self.ssl_verify = ssl_verify
        self.headers = {"Content-Type": "application/json"}
        self.base_url = f"https://{host}/{mode}/v{api_version}/"

    def login(self):
        """Authenticate and store session ID."""
        login_payload = {"user": self.username, "password": self.password}
        url = self.base_url + "login"
        response = requests.post(
            url, data=json.dumps(login_payload),
            headers=self.headers, verify=self.ssl_verify,
        )
        resp_struct = response.json()
        self.headers["X-chkp-sid"] = resp_struct["sid"]

    def call(self, endpoint, payload=None):
        """Make an API call."""
        url = self.base_url + endpoint
        if payload is None:
            payload = {}
        response = requests.post(
            url, data=json.dumps(payload),
            headers=self.headers, verify=self.ssl_verify
        )
        return response

    def logout(self):
        """Logout and remove session."""
        res = self.call("logout")
        if res.status_code == 200:
            self.headers.pop("X-chkp-sid", None)
```

PyNEng Style — Simple Device Class

```python
class NetworkDevice:
    def __init__(self, hostname, ip, vendor):
        self.hostname = hostname
        self.ip = ip
        self.vendor = vendor
        self.interfaces = []

    def add_interface(self, name, ip=None):
        self.interfaces.append({"name": name, "ip": ip})

    def show_info(self):
        print(f"Device: {self.hostname} ({self.ip}) - {self.vendor}")
        for intf in self.interfaces:
            print(f"  {intf['name']}: {intf.get('ip', 'no ip')}")

# Usage
rtr = NetworkDevice("core-rtr-01", "10.0.0.1", "Cisco")
rtr.add_interface("Gi0/0", "10.0.0.1")
rtr.add_interface("Gi0/1", "10.0.1.1")
rtr.show_info()
```

Chapter 17 Lab Tasks

Task 17.1: Create a `Router` class with attributes: hostname, ip, vendor, model. Add a `__init__` method and a `show_info()` method. Create 2 instances and call `show_info()` on each.

Task 17.2: Add an `add_interface()` method to your Router class that appends interface dicts to a list attribute. Add 3 interfaces and print the list.

Task 17.3: Create a `Firewall` class with a `login()` method that sets `self.authenticated = True` and a `logout()` method that sets it to `False`. Test the state transitions.

Task 17.4: Create an API client class with `__init__`, `login`, `call`, and `logout` methods (stubs that print messages). Chain them together in a test script.

Task 17.5: Create a class `VlanManager` that stores VLANs in a set. Add methods: `add_vlan()`, `remove_vlan()`, `list_vlans()`, and `common_vlans(other_manager)`.

Chapter 18: OOP — Special Methods and Inheritance

18.1 Special (Dunder) Methods

```python
class IPAddress:
    def __init__(self, ip):
        self.ip = ip
        self.octets = [int(o) for o in ip.split(".")]

    def __str__(self):
        return self.ip

    def __repr__(self):
        return f"IPAddress('{self.ip}')"

    def __eq__(self, other):
        return self.ip == other.ip

    def __lt__(self, other):
        return self.octets < other.octets

ip1 = IPAddress("10.0.0.1")
ip2 = IPAddress("10.0.0.2")
print(ip1)              # 10.0.0.1
print(repr(ip1))        # IPAddress('10.0.0.1')
print(ip1 == ip2)       # False
print(ip1 < ip2)        # True
```

18.2 Inheritance

```python
```

```python
class NetworkDevice:
    def __init__(self, hostname, ip):
        self.hostname = hostname
        self.ip = ip

    def connect(self):
        print(f"Connecting to {self.hostname} at {self.ip}")

class CiscoRouter(NetworkDevice):
    def __init__(self, hostname, ip, ios_version):
        super().__init__(hostname, ip)
        self.ios_version = ios_version
        self.vendor = "Cisco"

    def show_version(self):
        return f"{self.hostname} running IOS {self.ios_version}"

class AristaSwitch(NetworkDevice):
    def __init__(self, hostname, ip, eos_version):
        super().__init__(hostname, ip)
        self.eos_version = eos_version
        self.vendor = "Arista"

# Usage
rtr = CiscoRouter("rtr1", "10.0.0.1", "17.3.4a")
rtr.connect()                    # inherited from NetworkDevice
print(rtr.show_version())       # CiscoRouter method
```

Chapter 18 Lab Tasks

Task 18.1: Add `__str__` and `__repr__` methods to your Router class from Chapter 17. Print an instance with both `print()` and `repr()`.

Task 18.2: Create a base class `NetworkDevice` and two subclasses `Router` and `Switch`. Each subclass should add vendor-specific methods.

Task 18.3: Create an `IPAddress` class with `__eq__` and `__lt__` so IP addresses can be compared and sorted. Sort a list of 5 IP objects.

Task 18.4: Demonstrate `super().__init__()` by creating a `CiscoFirewall` class that inherits from `NetworkDevice` and adds firewall-specific attributes.

Task 18.5: Create a context manager class `DeviceConnection` with `__enter__` and `__exit__` methods that simulate connecting and disconnecting from a device.

Chapter 19: OS Interaction — subprocess, pathlib, os

19.1 subprocess

KByers Style

```python
#!/usr/bin/env python
import subprocess

# subprocess.run() — recommended approach
command = ["ping", "-c", "4", "google.com"]
result = subprocess.run(command, capture_output=True, text=True)

print(result.stdout)
print(result.stderr)
print(result.returncode)      # 0 = success
```

19.2 pathlib

KByers Style — Modern File Path Handling

```python
```

```python
from pathlib import Path
import yaml

home = Path.home()
fw_file = home / "python_course" / "firewalls.yml"

print(f"Exists: {fw_file.exists()}")
print(f"Is file: {fw_file.is_file()}")
print(f"Parent dir: {fw_file.parent}")

# Create directories
work_dir = home / "tmp_work"
if not work_dir.exists():
    work_dir.mkdir(parents=True, exist_ok=True)

# Find all JSON files recursively
course_dir = home / "python_course"
for f in course_dir.rglob("*.json"):
    print(f)
```

19.3 os.system() vs subprocess

```python
import os

# os.system() — simple but limited (no output capture)
os.system("ping -c 4 google.com")

# subprocess — full control (preferred)
import subprocess
result = subprocess.run(["ping", "-c", "4", "google.com"],
                        capture_output=True, text=True)
print(result.stdout)
```

Chapter 19 Lab Tasks

Task 19.1: Use `subprocess.run()` to execute `ping -c 3 8.8.8.8`. Print stdout, stderr, and the return code.

Task 19.2: Use `pathlib.Path` to check if `/etc/hosts` exists, print its parent directory, and read its first 5 lines.

Task 19.3: Use `pathlib` to find all `.py` files in your home directory (recursively). Count them and print the total.

Task 19.4: Write a function that takes a command as a list and uses `subprocess.Popen` to run it, returning stdout, stderr, and the return code.

Task 19.5: Use `pathlib` to create a directory structure: `~/lab/configs/`, `~/lab/logs/`, `~/lab/scripts/`. Verify each exists.

Chapter 20: Connecting to Network Devices (Netmiko)

20.1 First Connection

Netmiko Course Style

```python
import os
from getpass import getpass
from netmiko import ConnectHandler

password = os.getenv("NETMIKO_PASSWORD") if os.getenv("NETMIKO_PASSWORD") else getpa

# Simple connection
net_connect = ConnectHandler(
    device_type="cisco_ios",
    host="cisco3.lasthop.io",
    username="pyclass",
    password=password,
)
print(net_connect.find_prompt())
net_connect.disconnect()
```

20.2 Using a Device Dictionary

```python
```

```
my_device = {
    "device_type": "cisco_ios",
    "host": "cisco3.lasthop.io",
    "username": "pyclass",
    "password": password,
}

net_connect = ConnectHandler(**my_device)
print(net_connect.find_prompt())
net_connect.disconnect()
```

20.3 Context Manager (Auto-Disconnect)

```python
my_device = {
    "device_type": "cisco_ios",
    "host": "cisco3.lasthop.io",
    "username": "pyclass",
    "password": password,
}

with ConnectHandler(**my_device) as net_connect:
    print(net_connect.find_prompt())
# Automatically disconnected here
```

20.4 Session Logging

```python
my_device = {
    "device_type": "cisco_ios",
    "host": "cisco3.lasthop.io",
    "username": "pyclass",
    "password": password,
    "session_log": "cisco3.out",    # Logs everything to file
}

with ConnectHandler(**my_device) as net_connect:
    output = net_connect.send_command("show ip int brief")
    print(output)
```

20.5 Check Point Gaia Connections

KByers Style

```python
from netmiko import ConnectHandler

chkpt_fw = {
    "host": "chkpnt-pod99.lasthop.io",
    "device_type": "checkpoint_gaia",
    "username": "admin",
    "use_keys": True,
    "key_file": "/home/user/.ssh/eu-sshkey.pem",
    "session_log": "output.log",
}

with ConnectHandler(**chkpt_fw) as nc:
    print(nc.find_prompt())
    data = nc.send_command("show version all")
    print(data)
```

Chapter 20 Lab Tasks

Task 20.1: Write a Netmiko script that connects to a Cisco IOS device and prints the prompt. Use a device dictionary and context manager.

Task 20.2: Modify the script to enable session logging to a file. Verify the log file contents after running.

Task 20.3: Write a script that connects to an NX-OS device using a different `device_type`. Print the prompt.

Task 20.4: Create a function `connect_device(device_dict)` that handles the connection and returns the prompt. Handle `NetmikoAuthenticationException` and `NetmikoTimeoutException`.

Task 20.5: Write a script that reads device parameters from a YAML file and connects to each device, printing the prompt.

Chapter 21: Sending Commands with Netmiko

21.1 send_command()

```python
from netmiko import ConnectHandler

with ConnectHandler(**device) as net_connect:
    # send_command() waits for the prompt before returning
    output = net_connect.send_command("show ip interface brief")
    print(output)

    # With TextFSM parsing
    output = net_connect.send_command("show ip int brief", use_textfsm=True)
    print(output)    # Returns list of dicts!
```

21.2 send_command_timing()

```python
# send_command_timing() uses time-based approach (no prompt detection)
output = net_connect.send_command_timing("show ip route")
print(output)
```

21.3 write_channel() and read_channel()

Netmiko Course Style — Low-Level Control

```python
import time
from netmiko import ConnectHandler

with ConnectHandler(**my_device) as net_connect:
    # Send command directly to channel
    net_connect.write_channel("show ip int brief\n")
    time.sleep(1)
    output = net_connect.read_channel()
    print(output)
```

Chapter 21 Lab Tasks

Task 21.1: Connect to a device and use `send_command()` to gather `show version`, `show ip route`, and `show interfaces`. Print each output.

Task 21.2: Use `send_command()` with `use_textfsm=True` to parse `show ip int brief` into structured data. Loop through the results.

Task 21.3: Write a script that sends a command and saves the output to a text file named `{hostname}_{command}.txt`.

Task 21.4: Use `write_channel()` and `read_channel()` to manually interact with a device. Send "show clock" and read the response.

Task 21.5: Write a function that takes a device dict and a list of commands, connects, runs all commands, and returns a dictionary mapping each command to its output.

Chapter 22: Device Configuration with Netmiko

22.1 send_config_set()

```python
from netmiko import ConnectHandler

cfg_commands = [
    "logging buffered 20000",
    "logging console notifications",
    "no logging monitor",
]

with ConnectHandler(**device) as net_connect:
    output = net_connect.send_config_set(cfg_commands)
    print(output)
```

22.2 send_config_from_file()

```python
```

```
# File: changes.cfg
# interface Loopback99
#  ip address 10.99.99.1 255.255.255.255

with ConnectHandler(**device) as net_connect:
    output = net_connect.send_config_from_file("changes.cfg")
    print(output)
```

22.3 Enable Mode and Config Mode

Netmiko Course Style

```python
with ConnectHandler(**device) as net_connect:
    # Enable mode
    net_connect.enable()
    print(net_connect.find_prompt())    # rtr1#

    # Enter config mode manually
    net_connect.config_mode()
    print(net_connect.find_prompt())    # rtr1(config)#

    # Exit config mode
    net_connect.exit_config_mode()
```

22.4 Save Configuration

```python
with ConnectHandler(**device) as net_connect:
    net_connect.send_config_set(["interface Lo99", "ip address 10.99.99.1 255.255.25
    net_connect.save_config()       # write memory / copy run start
```

22.5 Juniper Commit Operations

Netmiko Course Style

```python
```

```python
vmx = {
    "device_type": "juniper_junos",
    "host": "vmx1.lasthop.io",
    "username": "pyclass",
    "password": password,
}

with ConnectHandler(**vmx) as net_connect:
    cfg_commands = [
        "set system syslog archive size 110k files 3",
        "set system time-zone America/New_York",
    ]
    output = net_connect.send_config_set(cfg_commands)
    print(output)

    # Commit with comment
    output = net_connect.commit(
        comment="Configuration change using Netmiko"
    )
    print(output)
```

Chapter 22 Lab Tasks

Task 22.1: Write a script that pushes 3 configuration commands to a Cisco IOS device using `send_config_set()`. Print the output.

Task 22.2: Create a config file with 5 interface commands. Use `send_config_from_file()` to apply them.

Task 22.3: Write a script that enters enable mode, verifies the privilege level, enters config mode, makes a change, exits, and saves.

Task 22.4: Write a reusable function `apply_config(device, commands)` that connects, pushes config, saves, and returns the output.

Task 22.5: Write a script for a Juniper device that sends configuration commands and commits with a comment. Verify the commit history.

Chapter 23: REST APIs with Python (requests)

23.1 GET Requests

```python
import requests

# Simple GET
response = requests.get("https://api.ipify.org?format=json")
print(response.status_code)    # 200
print(response.json())         # {'ip': '203.0.113.5'}
```

23.2 POST Requests with Authentication

KByers Style — Check Point API Authentication

```python
```

```python
import requests
import json
import os
from dotenv import load_dotenv

load_dotenv()
host = "chkpnt-pod99.lasthop.io"
base_url = f"https://{host}/gaia_api/v1.8/"
user = "admin"
password = os.environ["CHKP_ADMIN"]

# Login (POST)
url = f"{base_url}login"
headers = {"Content-Type": "application/json"}
login_payload = {"user": user, "password": password}

response = requests.post(
    url, data=json.dumps(login_payload),
    headers=headers, verify=False
)

session_id = response.json()["sid"]
headers["X-chkp-sid"] = session_id

# Make API calls with session
url = f"{base_url}show-api-versions"
response = requests.post(url, data=json.dumps({}), headers=headers, verify=False)
print(response.json())

# Logout
url = f"{base_url}logout"
response = requests.post(url, data=json.dumps({}), headers=headers, verify=False)
```

23.3 Environment Variables and .env Files

```python
```

```python
import os
from dotenv import load_dotenv

# .env file:
# CHKP_ADMIN=mysecretpassword
# CHKP_EXPERT=myexpertpassword

load_dotenv()
admin_pass = os.environ["CHKP_ADMIN"]
expert_pass = os.environ["CHKP_EXPERT"]
```

Chapter 23 Lab Tasks

Task 23.1: Use `requests.get()` to fetch data from a public API (e.g., https://api.ipify.org?format=json). Print the response status code and JSON body.

Task 23.2: Write a function `api_login(base_url, username, password)` that performs a POST login and returns the session headers. Include error handling.

Task 23.3: Write a function `api_call(base_url, endpoint, headers, payload)` that makes a POST API call and returns the response. Test it.

Task 23.4: Create a `.env` file with credentials and use `python-dotenv` to load them. Verify they are accessible via `os.environ`.

Task 23.5: Write a complete script that logs in to an API, makes 2 API calls, and logs out. Structure it with functions and `if __name__ == "__main__"`.

Chapter 24: Concurrent Connections — Threads and Processes

24.1 Why Concurrency?

Connecting to 100 devices sequentially takes minutes. Concurrency lets you connect to many devices simultaneously.

24.2 ThreadPoolExecutor

KByers Style

```python
```

```python
from concurrent.futures import ThreadPoolExecutor, as_completed
from datetime import datetime
from netmiko import ConnectHandler

def ssh_conn(device):
    net_connect = ConnectHandler(**device)
    return net_connect.find_prompt()

if __name__ == "__main__":
    start_time = datetime.now()
    max_threads = 4

    pool = ThreadPoolExecutor(max_threads)

    future_list = []
    for a_device in device_list:
        future = pool.submit(ssh_conn, a_device)
        future_list.append(future)

    for future in as_completed(future_list):
        print("Result: " + future.result())

    end_time = datetime.now()
    print(f"Elapsed time: {end_time - start_time}")
```

24.3 Context Manager Style

```python
from concurrent.futures import ThreadPoolExecutor, as_completed

with ThreadPoolExecutor(max_workers=4) as pool:
    future_list = []
    for device in device_list:
        future = pool.submit(ssh_conn, device)
        future_list.append(future)

    for future in as_completed(future_list):
        print("Result: " + future.result())
```

24.4 ProcessPoolExecutor

```python
from concurrent.futures import ProcessPoolExecutor, as_completed

# Same API, just swap ThreadPoolExecutor for ProcessPoolExecutor
with ProcessPoolExecutor(max_workers=4) as pool:
    future_list = []
    for device in device_list:
        future = pool.submit(ssh_conn, device)
        future_list.append(future)

    for future in as_completed(future_list):
        host, data = future.result()
        print(f"{host}: {data}")
```

24.5 Threads vs Processes

Feature	ThreadPoolExecutor	ProcessPoolExecutor
Best for	I/O bound (SSH, API calls)	CPU bound (parsing)
GIL impact	Limited by GIL	Bypasses GIL
Memory	Shared	Separate per process
Use case	Network automation	Data processing

Chapter 24 Lab Tasks

Task 24.1: Write a function `ssh_conn(device)` that simulates an SSH connection (sleep 1 second, return hostname). Use ThreadPoolExecutor with 5 threads to connect to 10 "devices".

Task 24.2: Measure the time difference between sequential execution (for loop) and concurrent execution (ThreadPoolExecutor) for 10 simulated connections.

Task 24.3: Use `as_completed()` to process results as they finish. Print each result with a timestamp showing the order of completion.

Task 24.4: Add error handling to your concurrent function: catch exceptions from individual futures and print error messages without crashing the whole batch.

Task 24.5: Write a concurrent script that reads devices from a YAML file, connects to each in parallel, collects `show version`, and writes results to a JSON file.

Chapter 25: Jinja2 Configuration Templates

25.1 Basic Templates

```python
from jinja2 import Environment, FileSystemLoader

env = Environment(loader=FileSystemLoader("templates"))
template = env.get_template("interface.j2")

# Template file: templates/interface.j2
# interface {{ interface_name }}
#  ip address {{ ip_address }} {{ subnet_mask }}
#  no shutdown

output = template.render(
    interface_name="GigabitEthernet0/1",
    ip_address="10.0.0.1",
    subnet_mask="255.255.255.0"
)
print(output)
```

25.2 Loops and Conditionals in Templates

```jinja2
```

```
{# templates/switch_config.j2 #}
hostname {{ hostname }}
!
{% for vlan in vlans %}
vlan {{ vlan.id }}
 name {{ vlan.name }}
{% endfor %}
!
{% for intf in interfaces %}
interface {{ intf.name }}
{% if intf.mode == "access" %}
 switchport mode access
 switchport access vlan {{ intf.vlan }}
{% elif intf.mode == "trunk" %}
 switchport mode trunk
 switchport trunk allowed vlan {{ intf.allowed_vlans }}
{% endif %}
 no shutdown
{% endfor %}
```

Chapter 25 Lab Tasks

Task 25.1: Create a Jinja2 template for a router interface config. Render it with 3 different sets of variables.

Task 25.2: Create a template with a for loop that generates VLAN configurations from a list of VLAN dicts.

Task 25.3: Add conditional logic to a template: if interface mode is "access", configure access mode; if "trunk", configure trunk.

Task 25.4: Write a script that reads device variables from a YAML file and renders a Jinja2 template for each device, writing output to separate files.

Task 25.5: Create a complete switch configuration template with hostname, VLANs, interfaces, and SNMP settings. Render it with realistic data.

Chapter 26: Parsing Output with TextFSM

26.1 Netmiko + TextFSM

```python
from netmiko import ConnectHandler

with ConnectHandler(**device) as net_connect:
    # Automatic TextFSM parsing
    output = net_connect.send_command(
        "show ip interface brief",
        use_textfsm=True
    )

    # Returns list of dicts
    for interface in output:
        print(f"{interface['intf']:25} {interface['ipaddr']:16} {interface['status']
```

26.2 TextFSM Template Syntax

```
# show_ip_int_brief.textfsm
Value INTF (\S+)
Value IPADDR (\S+)
Value STATUS (up|down|administratively down)
Value PROTO (up|down)

Start
  ^${INTF}\s+${IPADDR}\s+\w+\s+\w+\s+${STATUS}\s+${PROTO} -> Record
```

Chapter 26 Lab Tasks

Task 26.1: Use `send_command()` with `use_textfsm=True` to parse `show ip int brief` output. Print results as a formatted table.

Task 26.2: Write a custom TextFSM template to parse `show ip route` output. Extract network, mask, and next-hop.

Task 26.3: Parse `show interfaces` output to extract interface name, status, MTU, and bandwidth.

Task 26.4: Write a script that collects parsed data from 3 devices and combines results into a single report.

Task 26.5: Export TextFSM-parsed data to a CSV file for import into a spreadsheet.

Chapter 27: Testing with pytest

27.1 Basic Tests

```python
# test_network_utils.py
from network_utils import validate_ip, calculate_subnet_hosts

def test_valid_ip():
    assert validate_ip("10.0.0.1") == True

def test_invalid_ip():
    assert validate_ip("999.0.0.1") == False

def test_subnet_hosts_24():
    assert calculate_subnet_hosts(24) == 254

def test_subnet_hosts_30():
    assert calculate_subnet_hosts(30) == 2
```

Run: `pytest test_network_utils.py -v`

Chapter 27 Lab Tasks

Task 27.1: Write 3 pytest test functions for a `validate_ip()` function. Test valid IPs, invalid IPs, and edge cases.

Task 27.2: Write tests for a `generate_hostname()` function. Test default values and custom parameters.

Task 27.3: Use `pytest.raises()` to test that your custom exception is raised when expected.

Task 27.4: Create a test file with a `@pytest.fixture` that provides sample device data to multiple test functions.

Task 27.5: Write a test that verifies a function correctly reads a JSON file and returns the expected data structure. Use `tmp_path` fixture.

Chapter 28: Working with Databases (SQLite)

28.1 SQLite Basics

```python
import sqlite3

# Connect (creates file if not exists)
conn = sqlite3.connect("network_inventory.db")
cursor = conn.cursor()

# Create table
cursor.execute("""
    CREATE TABLE IF NOT EXISTS devices (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        hostname TEXT NOT NULL,
        ip TEXT NOT NULL,
        vendor TEXT,
        model TEXT,
        site TEXT
    )
""")

# Insert data
cursor.execute(
    "INSERT INTO devices (hostname, ip, vendor, model, site) VALUES (?, ?, ?, ?, ?)"
    ("rtr1", "10.0.0.1", "Cisco", "ISR4431", "NYC")
)
conn.commit()

# Query data
cursor.execute("SELECT hostname, ip, vendor FROM devices WHERE site = ?", ("NYC",))
rows = cursor.fetchall()
for row in rows:
    print(f"{row[0]:15} {row[1]:16} {row[2]}")

conn.close()
```

Chapter 28 Lab Tasks

Task 28.1: Create a SQLite database with a `devices` table. Insert 5 devices and query all of them.

Task 28.2: Write a function `add_device(conn, hostname, ip, vendor)` and `get_all_devices(conn)` to abstract database operations.

Task 28.3: Query the database for all Cisco devices and print a formatted table.

Task 28.4: Write a script that reads devices from a YAML file and inserts them into a SQLite database.

Task 28.5: Create a second table `interfaces` linked to `devices` via foreign key. Insert interface data and join the tables in a query.

Chapter 29: Building a Complete Network Automation Project

29.1 Project Structure

KByers Style — Real-World Check Point Automation

```
firewall_automation/
├── .env                        # Credentials (never commit to git!)
├── .gitignore                  # Excludes .env, *.pyc, __pycache__
├── requirements.txt            # pip dependencies
├── main.py                     # Entry point
├── config/
│   ├── devices.yaml            # Device inventory
│   └── blocked_ips.txt         # IP blocklist
├── lib/
│   ├── __init__.py
│   ├── chkpt_api.py            # API client class
│   ├── object_funcs.py         # Host/network object functions
│   ├── policy_funcs.py         # Firewall rule functions
│   └── exceptions.py           # Custom exceptions
├── templates/
│   └── switch_config.j2        # Jinja2 templates
└── tests/
    ├── test_api.py
    └── test_objects.py
```

29.2 Putting It All Together

```python
#!/usr/bin/env python
"""Main automation script — complete workflow."""
import os
from dotenv import load_dotenv
from lib.chkpt_api import ChkptAPI
from lib.object_funcs import cfg_host_objects
from lib.policy_funcs import cfg_fw_rules, install_fw_policy
from lib.exceptions import ChkPntConfigError

def main():
    load_dotenv()
    host = "chkpnt-pod99.lasthop.io"
    username = "admin"
    password = os.environ["CHKP_ADMIN"]

    api_client = ChkptAPI(host=host, username=username, password=password)

    try:
        api_client.login()
        print("[1/4] Configuring host objects...")
        cfg_host_objects(api_client, host_objects)
        print("[2/4] Configuring firewall rules...")
        cfg_fw_rules(api_client, fw_rules)
        print("[3/4] Publishing changes...")
        api_client.call("publish")
        print("[4/4] Installing policy...")
        install_fw_policy(api_client)
        print("Automation complete!")
    except ChkPntConfigError as e:
        print(f"Configuration failed: {e}")
    finally:
        api_client.logout()

if __name__ == "__main__":
    main()
```

Chapter 29 Lab Tasks

Task 29.1: Create a project directory structure as shown above. Initialize a git repo and create a `.gitignore` that excludes `.env`, `__pycache__`, and `*.pyc`.

Task 29.2: Write a `requirements.txt` with all packages needed for the project. Use `pip freeze` as a starting point, then clean it up.

Task 29.3: Create a complete automation script that: reads devices from YAML, connects via Netmiko, collects show commands, and writes results to JSON.

Task 29.4: Add error handling, logging (using `logging` module), and a retry mechanism to your script.

Task 29.5: Write pytest tests for at least 3 functions in your project. Run them and verify all pass.

Appendix A: Lab Task Answers

Chapter 1 Answers

1.1:

```bash
python3 -m venv ~/netauto
source ~/netauto/bin/activate
pip install netmiko requests pyyaml rich
pip list
```

1.2:

```python
# my_info.py
# This script displays my network engineer info
name = "John Smith"
role = "Network Engineer"
device_count = 150
print(f"Name: {name}")
print(f"Role: {role}")
print(f"Devices managed: {device_count}")
```

1.3:

```python
>>> 255 * 255        # 65025 (int)
>>> 10 / 3           # 3.3333... (float - division always returns float)
>>> 2 ** 32          # 4294967296 (int)
```

1.4:

```bash
mkdir ~/my_project && cd ~/my_project
git init
echo 'print("Hello!")' > hello.py
git add hello.py
git commit -m "first commit"
git log
```

1.5:

```python
from rich import print
print("[bold green]Router connected![/bold green]")
print("[bold red]Firewall blocked![/bold red]")
```

Chapter 2 Answers

2.1:

```python
hostname = "core-rtr-01"
ip_address = "10.10.10.1"
model = "ISR4431"
ios_version = "17.3.4a"
uptime_days = 365
print(f"Hostname: {hostname}")
print(f"IP: {ip_address}")
print(f"Model: {model}")
print(f"IOS: {ios_version}")
print(f"Uptime: {uptime_days} days")
```

2.2:

```python
for prefix in [24, 16, 8]:
    hosts = 2 ** (32 - prefix) - 2
    print(f"/{prefix}: {hosts} usable hosts")
# /24: 254, /16: 65534, /8: 16777214
```

2.3:

```python
print(bin(192))              # 0b11000000
print(hex(192))              # 0xc0
decimal = int("C0A80001", 16)
print(decimal)                # 3232235521
# That's 192.168.0.1 (C0=192, A8=168, 00=0, 01=1)
```

2.4:

```python
sent = 1000
received = 987
lost = sent - received
loss_pct = round((lost / sent) * 100, 2)
print(f"Packet loss: {loss_pct}%")   # 1.3%
```

2.5:

```python
bandwidth_gbps = 10.0
utilization = 0.73
used_gbps = bandwidth_gbps * utilization     # 7.3 Gbps
used_mbps = used_gbps * 1000                  # 7300 Mbps
bits_per_sec = used_gbps * 1_000_000_000      # 7,300,000,000 bps
bytes_in_60s = (bits_per_sec * 60) / 8
megabytes = bytes_in_60s / 1_000_000
print(f"Used: {used_mbps} Mbps")
print(f"Data in 60s: {megabytes:,.0f} MB")    # ~54,750 MB
```

Chapter 3 Answers

3.1:

```python
show_ip = "Interface Gi0/1, IP 192.168.1.1/24, Status up"
parts = show_ip.split(", ")
interface = parts[0].split(" ")[1]
ip_mask = parts[1].split(" ")[1]
ip_address, mask = ip_mask.split("/")
status = parts[2].split(" ")[1]
print(f"Interface: {interface}")
print(f"IP: {ip_address}")
print(f"Mask: /{mask}")
print(f"Status: {status}")
```

3.2:

```python
commands = [
    "interface GigabitEthernet0/1",
    " switchport mode access",
    " switchport access vlan 10",
    " spanning-tree portfast",
    " no shutdown"
]
config_block = "\n".join(commands)
print(config_block)
```

3.3:

```python
ipv6 = "2001:0db8:0000:0042:0000:0000:0000:0001"
hextets = ipv6.split(":")
compressed = [h.lstrip("0") or "0" for h in hextets]
result = ":".join(compressed)
print(result)    # 2001:db8:0:42:0:0:0:1
```

3.4:

```python
```

```
print(f"{'Hostname':<20} {'IP Address':<16} {'Status':^10}")
print(f"{'-'*20} {'-'*16} {'-'*10}")
print(f"{'core-rtr-01':<20} {'10.0.0.1':<16} {'up':^10}")
print(f"{'sw-access-01':<20} {'10.0.0.2':<16} {'up':^10}")
print(f"{'fw-edge-01':<20} {'10.0.0.3':<16} {'down':^10}")
```

3.5:

```python
intf = "GigabitEthernet0/0/0"
print(intf.startswith("Gig"))     # True
print("0/0" in intf)              # True
print(intf.endswith("/0"))        # True
```

Chapter 4 Answers

4.1:

```python
interfaces = [f"Gi0/{i}" for i in range(8)]
print(f"Third: {interfaces[2]}")
print(f"Last: {interfaces[-1]}")
print(f"Total: {len(interfaces)}")
```

4.2:

```python
vlans = [10, 20, 30, 40, 50, 60]
vlans.append(70);            print(vlans)
vlans.insert(0, 5);          print(vlans)
vlans.remove(30);            print(vlans)
vlans.sort();                print(vlans)
# Final: [5, 10, 20, 40, 50, 60, 70]
```

4.3:

```python
```

```
core = ["rtr1", "rtr2"]
access = ["sw1", "sw2", "sw3"]
all_devices_1 = core + access
all_devices_2 = core.copy()
all_devices_2.extend(access)
print(all_devices_1)
print(all_devices_2)
```

4.4:

```python
ips = ["10.0.0.1", "10.0.0.2", "10.0.0.3", "10.0.0.4", "10.0.0.5"]
first = ips.pop(0)
last = ips.pop()
print(f"Removed: {first}, {last}")
print(f"Remaining: {ips}")
```

4.5:

```python
config_lines = ["interface Gi0/1", "switchport mode access", "switchport access vlan
reversed_lines = config_lines[::-1]
print(reversed_lines)
```

Chapter 5 Answers

5.1:

```python
device = {
    "hostname": "rtr1", "ip": "10.0.0.1", "vendor": "Cisco",
    "model": "ISR4431", "os_version": "17.3.4a", "site": "NYC"
}
for key in device:
    print(f"{key}: {device[key]} | get: {device.get(key)}")
```

5.2:

```python
fw_rule = {
    "name": "Allow_Web",
    "source": {"type": "host", "ip": "10.0.0.5"},
    "destination": {"type": "network", "ip": "192.168.1.0", "mask": "/24"},
    "action": "permit"
}
dest_ip = fw_rule["destination"]["ip"]
print(f"Destination: {dest_ip}")    # 192.168.1.0
```

5.3:

```python
rtr1 = {"hostname": "rtr1", "ip": "10.0.0.1", "vendor": "Cisco"}
rtr2 = {"hostname": "rtr2", "ip": "10.0.0.2", "vendor": "Juniper"}
inventory = {rtr1["hostname"]: rtr1, rtr2["hostname"]: rtr2}
print(inventory)
```

5.4:

```python
device = {"hostname": "rtr1", "ip": "10.0.0.1", "vendor": "Cisco", "model": "ISR4431
for key, value in device.items():
    print(f"{key:>12}: {value}")
```

5.5:

```python
device = {"device_type": "arista_eos", "host": "10.0.0.1", "username": "admin", "pas
device["session_log"] = "arista.log"
device["device_type"] = "arista_eos_ssh"
device.pop("session_log")
print(device)
```

Chapter 6 Answers

6.1:

```python
dns = ("8.8.8.8", "8.8.4.4", "1.1.1.1")
# dns[0] = "9.9.9.9"  # TypeError: 'tuple' object does not support item assignment
primary, secondary, tertiary = dns
print(f"Primary: {primary}, Secondary: {secondary}, Tertiary: {tertiary}")
```

6.2:

```python
sw1 = {10, 20, 30, 40, 50}
sw2 = {30, 40, 50, 60, 70}
print(f"All: {sw1 | sw2}")
print(f"Common: {sw1 & sw2}")
print(f"Only sw1: {sw1 - sw2}")
print(f"Unique to one: {sw1 ^ sw2}")
```

6.3:

```python
dupes = [10, 20, 30, 10, 20, 40]
unique_set = set(dupes)
print(unique_set)                    # {10, 20, 30, 40}
sorted_list = sorted(unique_set)
print(sorted_list)                   # [10, 20, 30, 40]
```

6.4:

```python
values = [0, "", [], {}, None, "hello", [1], 42]
for v in values:
    print(f"{str(v):>10} -> {bool(v)}")
```

6.5:

```python
```

```python
current = {"permit_web", "permit_ssh", "deny_telnet"}
desired = {"permit_web", "permit_ssh", "permit_dns"}
to_add = desired - current
to_remove = current - desired
print(f"Add: {to_add}")        # {'permit_dns'}
print(f"Remove: {to_remove}")  # {'deny_telnet'}
```

Chapters 7-29 Answers (Selected Key Solutions)

7.1:

```python
vlan = int(input("Enter VLAN: "))
if vlan == 1:
    print("Default VLAN — do not use")
elif 2 <= vlan <= 1000:
    print("Standard VLAN")
elif 1001 <= vlan <= 4094:
    print("Extended VLAN")
else:
    print("Invalid VLAN")
```

8.1:

```python
ips = [f"10.0.0.{i}" for i in range(1, 11)]
for idx, ip in enumerate(ips, start=1):
    print(f"{idx}. {ip}")
```

9.1:

```python
def generate_hostname(site, role, number="01"):
    return f"{site}-{role}-{number}"

print(generate_hostname("NYC", "RTR"))         # NYC-RTR-01
print(generate_hostname("LAX", "SW", "05"))    # LAX-SW-05
```

12.3:

python

```python
class InvalidVlanError(Exception):
    pass

def validate_vlan(vlan_id):
    if not 1 <= vlan_id <= 4094:
        raise InvalidVlanError(f"VLAN {vlan_id} is outside valid range 1-4094")
    return True

try:
    validate_vlan(5000)
except InvalidVlanError as e:
    print(e)
```

16.1:

python

```python
ips = [f"192.168.1.{i}" for i in range(1, 21)]
print(ips)
```

17.1:

python

```python
class Router:
    def __init__(self, hostname, ip, vendor, model):
        self.hostname = hostname
        self.ip = ip
        self.vendor = vendor
        self.model = model

    def show_info(self):
        print(f"{self.hostname} ({self.ip}) - {self.vendor} {self.model}")

rtr1 = Router("core-rtr-01", "10.0.0.1", "Cisco", "ISR4431")
rtr2 = Router("edge-rtr-01", "10.0.0.2", "Juniper", "MX204")
rtr1.show_info()
rtr2.show_info()
```

20.1:

python

```python
from netmiko import ConnectHandler
from getpass import getpass

device = {
    "device_type": "cisco_ios",
    "host": "192.168.1.1",
    "username": "admin",
    "password": getpass(),
}

with ConnectHandler(**device) as nc:
    print(nc.find_prompt())
```

24.1:

```python
from concurrent.futures import ThreadPoolExecutor, as_completed
import time

def ssh_conn(device):
    time.sleep(1)
    return f"Connected to {device['hostname']}"

devices = [{"hostname": f"rtr{i}"} for i in range(1, 11)]

with ThreadPoolExecutor(max_workers=5) as pool:
    futures = [pool.submit(ssh_conn, d) for d in devices]
    for future in as_completed(futures):
        print(future.result())
```

29.3:

```python
```

```python
import yaml
import json
from netmiko import ConnectHandler

with open("devices.yaml") as f:
    devices = yaml.safe_load(f)

results = {}
for device in devices:
    hostname = device["hostname"]
    with ConnectHandler(**device) as nc:
        results[hostname] = {
            "prompt": nc.find_prompt(),
            "version": nc.send_command("show version"),
        }

with open("results.json", "w") as f:
    json.dump(results, f, indent=2)

print(f"Collected data from {len(results)} devices")
```

End of Lab Guide

Sources: LPTHW (Zed Shaw), PyNEng (Natasha Samoylenko), Python Course & Netmiko Course (Kirk Byers)

Total Chapters: 29 + Appendix | **Total Lab Tasks:** 145 with answers